CHOKEPOINT

STRAIT OF HORMUZ SCENARIO BRIEF · ANALYTICA DATA SCIENCE SOLUTIONS

data as of **2026-07-22**
chokepoint.analyticadss.com · 100% public data

Hormuz disruption: 100% — “Full closure”

Bypass assumption: Conservative — EIA spare bypass. Every figure carries {source, url, retrieved, status} in the published dataset. Five pages: exposure · markets · history & structure · the 2026 ladder · method.

18.3 M b/d
STRANDED — NOWHERE TO GO

20.9 M b/d
GROSS DISRUPTED — THE HEADLINE NUMBER

2.6/2.6
BYPASS ABSORBING, M B/D

+117–220%
BRENT SHOCK BAND — ILLUSTRATIVE

66 days
STRATEGIC-RESERVE RUNWAY

+50%
SIEGE PREMIUM, 12–24 MO — PERSISTENT WHILE BARRELS STAY MISSING

17.0 M b/d
NET GAP AFTER MAX SPR RELEASE — THE TAP BINDS AT 1.3 M B/D

83 Mt/yr
LNG OFFLINE, ZERO BYPASS — GAS MULTIPLIER ×3.6–×6.2 (ILLUSTR.)

1-in-13–100
IMPLIED HULL-LOSS ODDS PER TRANSIT, FROM CURRENT QUOTES

Still transiting **0.0** · Rerouted overland **2.6** · Stranded **18.3** — of **20.9** M b/d pre-crisis throughput (EIA, H1 2025) · stranded = **17.6%** of world supply

PRODUCER	EXPORTS	STILL MOVING VS STRANDED	STRANDED	EXPOSURE
Saudi Arabia	5.5		−4.0	PARTIAL
Iraq	3.2		−3.2	HIGH
Kuwait	1.4		−1.4	TOTAL
Qatar	1.3		−1.3	TOTAL
Iran	1.5		−1.2	HIGH
United Arab Emirates	1.9		−1.1	PARTIAL
Bahrain	0.2		−0.2	HIGH

Gulf seaborne exports, M b/d, pre-crisis baselines (EIA country briefs). Per producer: stranded = exports·d − min(own pipeline spare, exports·d). Only Saudi Arabia (Petroline → Yanbu), the UAE (ADCOP → Fujairah) and Iran (Goreh–Jask) own an overland escape.

> **The number nobody states cleanly:** Qatar's ~77 Mt/yr of LNG — about a fifth of the world's LNG trade — has zero overland bypass. Petroline can save Saudi barrels; nothing can save Qatari cargoes.

HOW TO READ THIS

Gross vs. net is the whole argument. Headlines quote gross throughput disrupted; the honest damage number subtracts spare overland bypass — capacity that exists **and isn't already full**. EIA's conservative spare figure is 2.6 M b/d; post-2025 Petroline-expansion commentary supports up to 5.5, so it ships as a switchable assumption, not a footnote.

The price band is **illustrative by construction** — real markets price expectations, OPEC spare capacity and SPR releases. History's verdict from 45 years of Hormuz threats (p. 4): spikes mean-revert unless barrels are physically missing. This time they are.

WAR-RISK CONTEXT

Hull war-risk cover for a Gulf transit: **1–7.5%** of vessel value (2026-07) vs **0.25%** pre-crisis — for a $150M VLCC, **$1.5M–$11.3M per voyage**, when cover is offered at all. Full ledger and voyage arithmetic on p. 3.

WHY HORMUZ IS DIFFERENT

The only major oil chokepoint with **no sea alternative**. Suez closed for eight years and the world rerouted around the Cape; here the ships have nowhere else to go (comparison, p. 4).

Importer exposure — who hurts first

An importer's loss = its Hormuz volume × d × (stranded/gross); bypass relief is pro-rata because rerouted barrels are fungible. Days of cover = strategic stocks ÷ lost supply. Sorted shortest first.

IMPORTER	VIA HORMUZ, M B/D	LOSS, M B/D	% OF ITS DEMAND	STOCKS, M BBL	DAYS OF COVER
Other Asia & Pacific ESTIMATE	6.5	−5.7	63%	90	16
India	2.1	−1.8	33%	39	21
China ESTIMATE	5.4	−4.7	29%	400	85
Rest of world ESTIMATE	0.4	−0.4	–	50	143
South Korea	1.7	−1.5	53%	240	161
Europe (OECD)	2.8	−2.5	18%	570	232
Japan	1.6	−1.4	42%	470	335
United States	0.4	−0.4	2%	403	999+

Volumes: EIA WOTC destinations (2024) — they sum to total throughput, so importer losses sum exactly to stranded barrels (build-enforced). Stocks: IEA / ISPRL / DOE; China's are state-opaque — public analyst estimates only. The asymmetry is the story: Asia holds weeks; the OECD holds a year.

RESERVES — THE TAP, NOT THE TANK REPORTED

The stock/flow runway says **66 days** — but coordinated releases max out near **1.3 M b/d** (2022 precedent: 240 M bbl over ~6 months). At today's stranded rate that covers **7%** of the gap, leaving **17.0 M b/d** uncovered. At maximum drawdown the stocks last **923 days** — the tank outlasts the crisis; the tap is the constraint.

MACRO PASS-THROUGH ILLUSTRATIVE

Public rules of thumb (+10% oil ≈ +0.2–0.4 pp CPI, −0.1–0.2 pp GDP over a year) applied to the shock band: consumer prices **+2.3–8.8 pp**; global growth **−1.2–4.4 pp**. A 2-pp global drag ≈ a typical recession's demand loss. Linear rules break down at extreme shocks — treat upper ends as directional. (IMF WEO; Fed staff ranges.)

THE LNG SHOCK — NO ESCAPE EXISTS ILLUSTRATIVE

83 Mt/yr
LNG OFFLINE (11.0 BCF/D)

20.6%
OF GLOBAL LNG TRADE

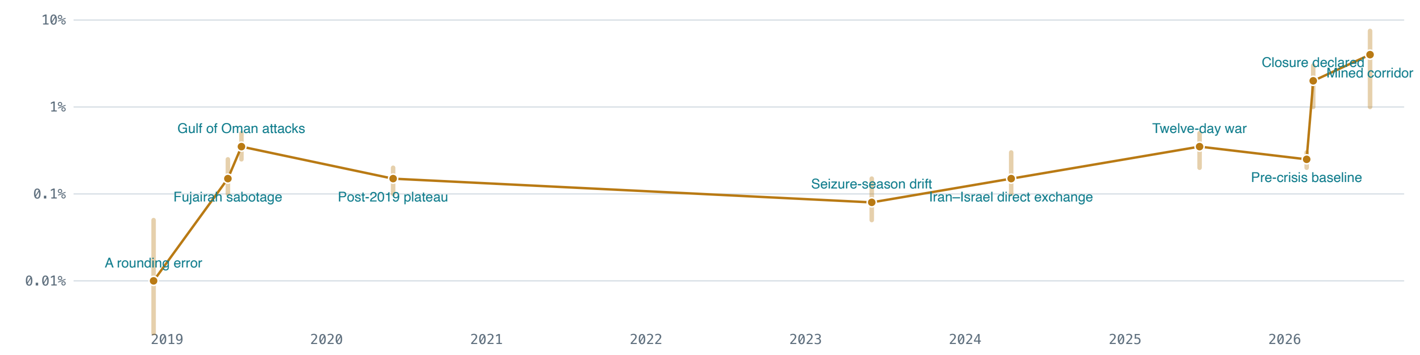
×3.6–×6.2
TTF/JKM MULTIPLIER BAND

BUYER	MT/YR VIA HORMUZ	LOST AT 100%
Asia (China, India, Korea, Japan…)	60	−60
Europe	16	−16
Other	7	−7

There is no strategic reserve for gas. Europe's storage (~100 bcm, ~25% of annual demand) is a seasonal buffer measured in weeks of winter drawdown; Asian LNG buyers hold days, not months.

The empirical anchor: Russia's 2021–22 pipeline cuts removed roughly the gas-equivalent of 14% of globally traded LNG from Europe's supply — TTF peaked near 10× its pre-crisis level (≈€340/MWh in August 2022 vs ≈€35 in early 2021) before demand destruction bit. VERIFIED

War-risk insurance ledger — % of hull value per Gulf transit (log scale)



JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary. Points inside quoted ranges; bars show the public low–high. From a 0.01% plumbing fee to the number that decides whether a ship sails. Note the ratchet: each crisis's "normal" settles above the last one's.

1-in-13 – 1-in-100
IMPLIED ODDS OF HULL LOSS PER TRANSIT, CURRENT QUOTES

~1-in-10,000
SAME ARITHMETIC, DEC 2018

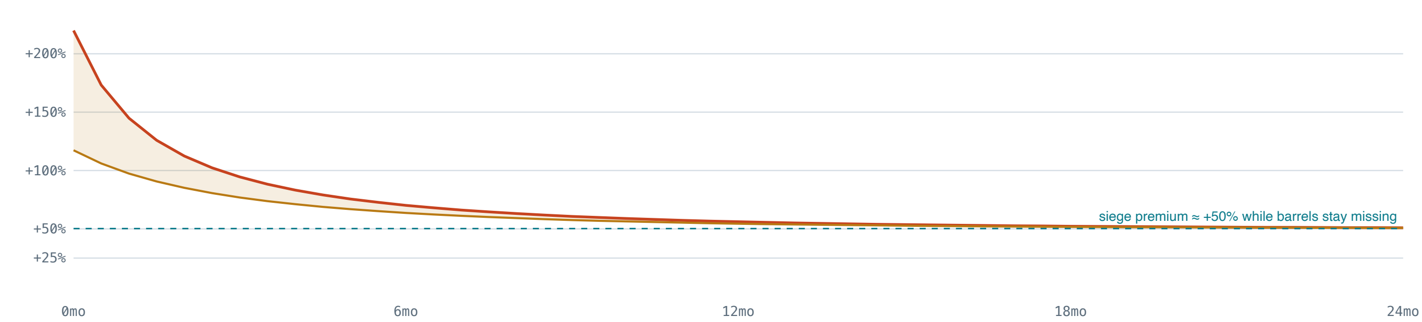
PREMIUM ≈ PROBABILITY × SEVERITY + LOADINGS → UPPER BOUNDS; AT 50% SEVERITY, DOUBLE THE 1-IN-N

Voyage economics — does the transit clear?

PREMIUM REGIME	% OF HULL	COST / TRANSIT	$ / BBL	SHARE OF FREIGHT	VERDICT
Feb 2026 “pre-crisis”	0.25%	$0.3M	$0.15	4%	CLEARS
Current — safer vessels	1%	$1.2M	$0.60	15%	CLEARS
Current — mid	4%	$4.8M	$2.40	60%	MARGINAL
Current — high-risk	7.5%	$9.0M	$4.50	113%	UNECONOMIC

$120M VLCC, 2 M bbl cargo, $4.0/bbl freight (public broker commentary; crisis chartering has spiked rates 3×+). One-way transit; round trips can double the premium. Above ~100%, the underwriter — not the charterer — decides whether the ship sails.

Dynamic price path — the spike decays; the siege premium stays



$\varepsilon(t) = \varepsilon_0 + (\varepsilon_\infty - \varepsilon_0)(1 - e^{-t/\tau})$, $\varepsilon_\infty = 0.35$, $\tau = 6$ months ILLUSTRATIVE — assumes the disruption persists; expectations, SPR releases and GDP feedback excluded. Substitution and demand destruction tame the spike, but converge to a persistent premium — not to zero — while barrels stay missing.

Event study — Brent after Hormuz threats (1/5/30 trading days)

EVENT	DATE	+1D	+5D	+30D
Tanker War escalates off Kharg ESTIMATE	1984-05-24	+1.0%	+2.0%	−4.0%
Bridgeton mine strike VERIFIED	1987-07-24	−2.8%	+0.2%	−10.2%
Operation Praying Mantis VERIFIED	1988-04-18	+3.6%	+1.8%	−1.5%
Iran closure threat (sanctions) VERIFIED	2011-12-27	−1.6%	+3.7%	+1.1%
Fujairah sabotage VERIFIED	2019-05-12	+1.0%	+3.2%	−11.3%
Gulf of Oman attacks VERIFIED	2019-06-13	+2.6%	+1.9%	+9.7%
Abqaiq–Khurais strike VERIFIED	2019-09-14	+11.7%	+6.5%	−4.0%
Soleimani strike VERIFIED	2020-01-03	+3.0%	−0.7%	−13.8%
Iran strikes Israel directly VERIFIED	2024-04-13	−2.4%	−5.5%	−13.0%
Israel–Iran twelve-day war VERIFIED	2025-06-13	+7.3%	+13.5%	+1.7%
2026 Hormuz crisis begins VERIFIED	2026-02-28	+8.3%	+34.2%	+70.9%

EIA Europe Brent spot (RBRTE) via FRED public mirror DCOILBRENTEU, daily; windows computed by the fetch pipeline (pre-1987 rows predate the daily series and remain estimates). The pattern: threat spikes mean-revert within weeks — even Abqaiq, the largest single supply disruption in oil-market history, round-tripped in a month — **unless barrels stay missing**. The 2026 row is the counterexample that proves the mechanism: +71% at 30 days and climbing.

Chokepoint anatomy — why Hormuz is different

CHOKEPOINT	OIL, M B/D	SEA ALTERNATIVE	BYPASS, M B/D	CLOSURE RECORD
Strait of Hormuz	20.9	NONE — NO WAY OUT BY SEA	2.6	Never fully closed through 45 years of threats — until 2026. The first actual closure is the current crisis.
Strait of Malacca	23.7	Sunda / Lombok straits +1–3d	0.4	Never closed; piracy-era harassment at worst. Rerouting costs days, not barrels.
Suez Canal + SUMED	9.2	Cape of Good Hope +10–14d	2.5	Closed 1956–57 and 1967–1975 (eight years); Ever Given 2021 (six days). The world rerouted and lived.
Bab el-Mandeb	8.6	Cape of Good Hope +10–14d	–	2023–25 Houthi campaign roughly halved transits — traffic diverted, cargoes still arrived.
Panama Canal	0.9	Cape Horn / Suez routings +5–13d	–	2023–24 drought cut transits ~30% — products rerouted at a freight cost.

EIA WOTC reference volumes. Every other chokepoint has an escape — a longer sea lane, a parallel pipeline, or both. Suez closed for eight years and the world rerouted. Hormuz's first real closure is a different kind of event.

Reopening runway — why d→0 is not day zero

FRICTION	PERSISTENCE	ANCHOR
Price premium	2–6 weeks	Threat premiums mean-revert within weeks once barrels flow — unless barrels stay missing. Abqaiq 2019: +14.6% one-day spike fully round-tripped in under a month despite 5.7 M b/d offline. VERIFIED
War-risk insurance	4–18 months	Underwriters normalize on renewal cycles and JWC reviews, not headlines — premiums stay multiples of the old baseline for months to years. After the 2019 spiral, Gulf rates held ~10× their April-2019 level well into 2021. ESTIMATE
Mined corridor	3–9 months	A swept convoy corridor takes months; certified full clearance takes years. Mines outlast every ceasefire that laid them. 1991 Gulf War: 1,200+ mines; coalition clearance operations ran for over two years. VERIFIED

The road to closure — 2025 → 2026

2025-06-13
Israel strikes Iran — the twelve-day war VERIFIED
Strikes on nuclear and military targets; Iran answers with missile barrages. Tankers begin GPS-jammed, zig-zag transits.
Flows: **Transits continue; loadings dip** · Price: **Brent +~7% day one**

2025-06-22
US strikes Fordow; Majlis votes to close the strait VERIFIED
B-2s hit Fordow, Natanz, Isfahan. Iran's parliament votes (non-binding) to close Hormuz — the decision rests with the SNSC, which doesn't act. The strait stays open.
Flows: **Self-deterrence: Iran's own exports transit the same water** · Price: **Brent touches ~$81 then collapses –7% on ceasefire news (24 Jun)**

2026-02-28
The 2026 crisis begins REPORTED
US and Israeli strikes on Iranian leadership and military targets. Iran's response makes 45 years of closure threats operational for the first time.
Flows: **Immediate transit hesitancy; insurers pause quotes** · Price: **+8.3% day one, +34% in a week, +71% in a month (daily spot series) — the first Hormuz spike that did not mean-revert**

2026-03-02
IRGC declares the strait closed REPORTED
VHF broadcasts warn 'no ship is allowed to pass'; transit permitted only for Iran-approved vessels. Mining of the central corridor and attacks on transiting ships follow.
Flows: **Tanker transits collapse toward –70%** · Price: **War-risk quotes move from 0.25% to 1–3% of hull within a week**

2026-04-11
US mine-clearing operations begin REPORTED
Two destroyers transit to 'set conditions' for de-mining; blockade operations against Iranian naval logistics run April–May.
Flows: **Escorted convoys only; ~150+ vessels damaged or on fire to date (reported)** · Price: **High-risk hull quotes reach 7.5–10%**

2026-06-15
Persian Gulf Strait Authority stands up ESTIMATE
Multinational escort-and-transit authority begins organizing convoys through swept corridors. ~80 mines still reported in the central corridor as of mid-July.
Flows: **Partial, escorted resumption; ~70% of normal transits still missing** · Price: **Premiums stay 4×–30× pre-crisis**

THE MODEL

```
stranded(d)   = T·d − min(spare_bypass, T·d)
ΔP/P          ≈ share / ε,  ε ∈ [0.08, 0.15], capped
price path    : ε(t) = ε₀ + (ε∞−ε₀)(1−e^(−t/τ))
SPR offset    = min(max_release_rate, stranded)
producer_i    : exports_i·d − min(pipeline spare_i, ·)
importer_i    : imports_i·d · stranded/gross
LNG mult      : 1 + (lng_lost/global_trade) / ε_gas
implied odds  : P(loss) ≤ premium / severity
voyage        : share = hull·p / (cargo·freight)
macro         : (ΔP%/10)·[coeffs], one-year
```

PROVENANCE LEGEND

verified — checked against the named public source at build time · **reported** — live-crisis public reporting · **estimate** — derived/approximate · **illustrative** — model assumption · **brief-baseline** — pending a fresh primary-source check. If a figure is an estimate, it says so.

SOURCES

EIA World Oil Transit Chokepoints (retrieved 2026-07-22); EIA country briefs, STEO and daily Brent · IEA oil security & gas market reports; GIIGNL · ISPRL, DOE, KNOC/METI stock disclosures · JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary · CRS R45281; US Navy histories · 2026-crisis public reporting · IMF WEO / Fed staff rules of thumb · Natural Earth. Per-figure citations with URLs and retrieval dates ship in the dataset and render in the site's Methodology panel.

INTENDED USE

Decision-support for shipping, energy and policy analysis from 100% public data. No real-time vessel positions, no navigational data, nothing that could serve as a targeting aid. Incident positions are approximate; vessel-behavior content is pattern-level.

Interactive version, all scenarios: **chokepoint.analyticadss.com**
Scenario deep link: http://localhost:5199/?intro=skip&brief=1#v=map&d=100&b=eia&y=2026&e=11
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